# Data Collection and Preprocessing Phase

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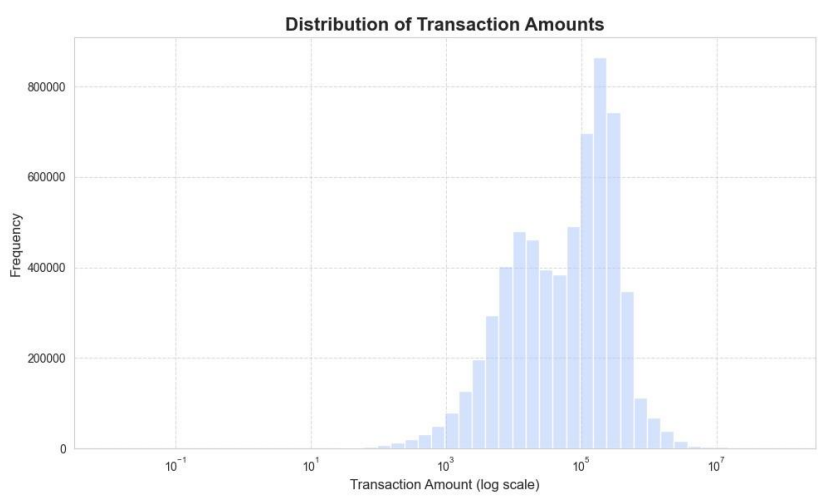
| Date | 02 Feb 2026 |
| Team ID | LTVIP2026TMIDS88548 |
| Project Title | Online Payment Fraud Detection using ML |
| Maximum Marks | 6 Marks |

## Data Exploration and Preprocessing

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

| Section | Description |
|---|---|
| Data Overview | |
| Univariate Analysis | |

| | step | amount | oldbalanceOrg | newbalanceOrig | oldbalanceDest | newbalanceDest | isFraud | isFlaggedFraud |
|---|---|---|---|---|---|---|---|---|
| count | 6.362620e+06 | 6.362620e+06 | 6.362620e+06 | 6.362620e+06 | 6.362620e+06 | 6.362620e+06 | 6.362620e+06 | 6.362620e+06 |
| mean | 2.433972e+02 | 1.798619e+05 | 8.338831e+05 | 8.551137e+05 | 1.100702e+06 | 1.224996e+06 | 1.290820e-03 | 2.514687e-06 |
| std | 1.423320e+02 | 6.038582e+05 | 2.888243e+06 | 2.924049e+06 | 3.399180e+06 | 3.674129e+06 | 3.590480e-02 | 1.585775e-03 |
| min | 1.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 25% | 1.560000e+02 | 1.338957e+04 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 |
| 50% | 2.390000e+02 | 7.487194e+04 | 1.420800e+04 | 0.000000e+00 | 1.327057e+05 | 2.146614e+05 | 0.000000e+00 | 0.000000e+00 |
| 75% | 3.350000e+02 | 2.087215e+05 | 1.073152e+05 | 1.442584e+05 | 9.430367e+05 | 1.111909e+06 | 0.000000e+00 | 0.000000e+00 |
| max | 7.430000e+02 | 9.244552e+07 | 5.958504e+07 | 4.958504e+07 | 3.560159e+08 | 3.561793e+08 | 1.000000e+00 | 1.000000e+00 |



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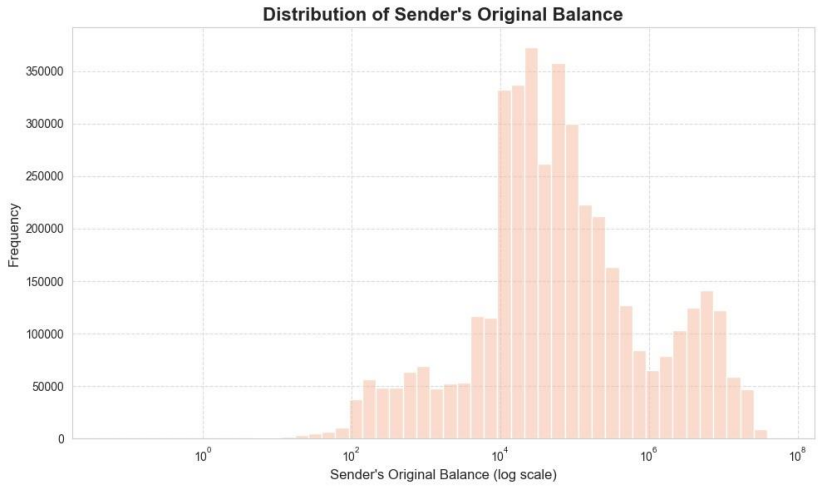
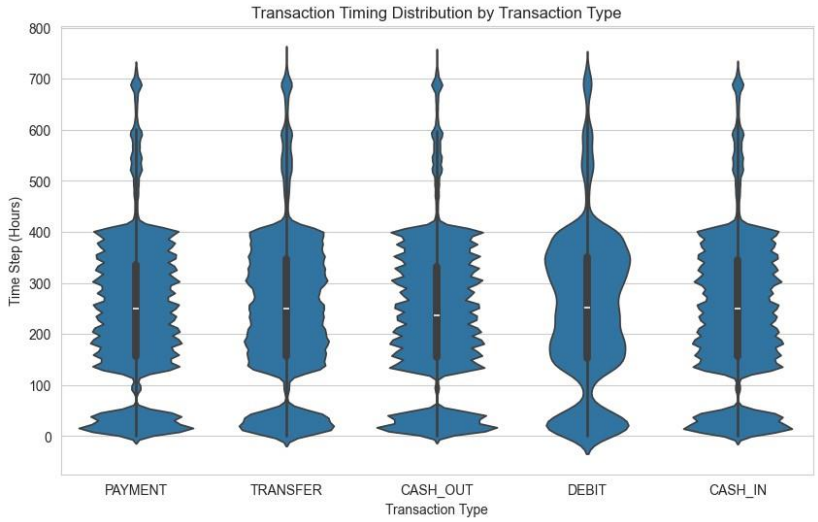
|---|---|
| |  |
| Bivariate Analysis |  |

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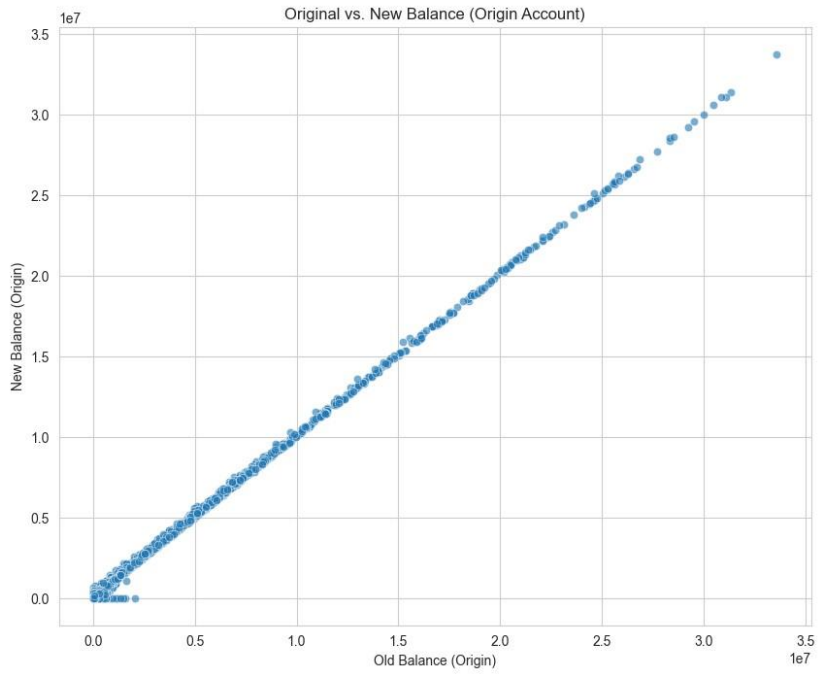
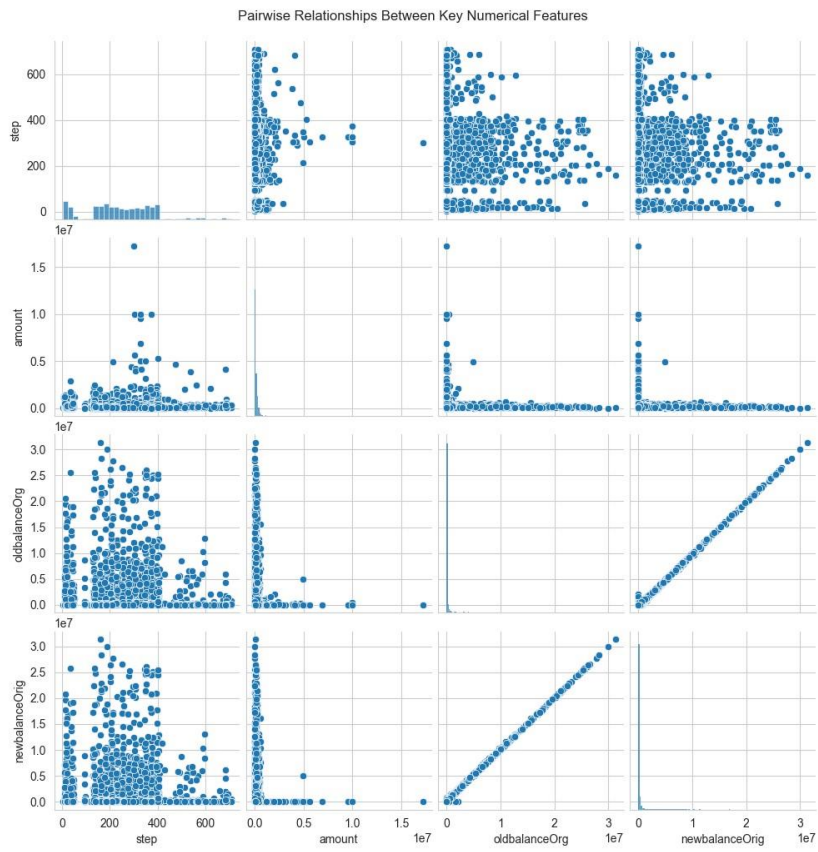
|---|---|
| |  |
| Multivariate Analysis |  |

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| Outliers and Anomalies | Due to the extremely wide distribution and heavy-tailed nature of financial transaction data in this dataset, traditional outlier detection methods are not effective. The data spans several orders of magnitude (e.g., transaction amounts range from $0 to $92+ million), making it difficult to distinguish between legitimate large transactions and true anomalies using standard statistical methods. |

**Data Preprocessing Code Screenshots**

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|---|---|
| Loading Data | |

```
#Loading Data
data=pd.read_csv("data.csv")
data_og=data.copy()
data.head()
```

| | step | type | amount | nameOrig | oldbalanceOrg | newbalanceOrig | nameDest | oldbalanceDest | newbalanceDest | isFraud | isFlaggedFraud |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | PAYMENT | 9839.64 | C1231006815 | 170136.0 | 160296.36 | M1979787155 | 0.0 | 0.0 | 0 | 0 |
| 1 | 1 | PAYMENT | 1864.28 | C1666544295 | 21249.0 | 19384.72 | M2044282225 | 0.0 | 0.0 | 0 | 0 |
| 2 | 1 | TRANSFER | 181.00 | C1305486145 | 181.0 | 0.00 | C553264065 | 0.0 | 0.0 | 1 | 0 |
| 3 | 1 | CASH_OUT | 181.00 | C840083671 | 181.0 | 0.00 | C38997010 | 21182.0 | 0.0 | 1 | 0 |
| 4 | 1 | PAYMENT | 11668.14 | C2048537720 | 41554.0 | 29885.86 | M1230701703 | 0.0 | 0.0 | 0 | 0 |

Handling Missing Data

```
#Handling Missing Data (There isnt any missing data)
data.isnull().sum()
```

```
step              0
type              0
amount            0
nameOrig          0
oldbalanceOrg     0
newbalanceOrig    0
nameDest          0
oldbalanceDest    0
newbalanceDest    0
isFraud           0
isFlaggedFraud    0
dtype: int64
```

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| Data Transformation | `df_clean = data.drop(['nameOrig', 'nameDest', 'isFlaggedFraud'], axis=1)`<br>`df_clean.columns.tolist()`<br><br>`['step', 'type', 'amount', 'oldbalanceOrg', 'newbalanceOrig', 'oldbalanceDest', 'newbalanceDest', 'isFraud']` |

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|---|---|
| | `df_encoded = pd.get_dummies(df_clean, columns=['type'], prefix='type')`<br>`df_encoded.head()`<br><br>amount, oldbalanceOrg, newbalanceOrig, oldbalanceDest, newbalanceDest, isFraud, type_CASH_IN, type_CASH_OUT, type_DEBIT, type_PAYMENT, type_TRANSFER<br>9839.64, 170136.0, 160296.36, 0.0, 0.0, 0, False, False, False, True, False<br>1864.28, 21249.0, 19384.72, 0.0, 0.0, 0, False, False, False, True, False<br>181.00, 181.0, 0.00, 0.0, 0.0, 1, False, False, False, False, True<br>181.00, 181.0, 0.00, 21182.0, 0.0, 1, False, True, False, False, False<br>11668.14, 41554.0, 29885.86, 0.0, 0.0, 0, False, False, False, True, False<br><br>`X = df_encoded.drop('isFraud', axis=1)`<br>`y = df_encoded['isFraud']`<br>`scaler = StandardScaler()`<br>`X_scaled = scaler.fit_transform(X)` |
| Feature Engineering | New features such as balance changes and ratios were considered but not used in the baseline due to time/resource constraints. |
| Save Processed Data | `X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)`<br>`X_scaled_df.describe()` |